

Mariam Mohammed Mahmoud Abdelgawad

AI Intern at FlyRank | Computer Science & Engineering Student

0106 583 0425 | 0120 437 2132 | marym2082006@gmail.com | El Bitash El Dekheila, Alexandria, Egypt
linkedin.com/in/mariam-abdelgawad-936436411 | mariam-abdelgawad.github.io/ai-portfolio

PROFESSIONAL SUMMARY

Third-year Computer Science & Engineering student at Egypt-Japan University of Science and Technology (E-JUST), CGPA 3.47/4.00. Built and documented five engineering and AI projects — including a 6-person IoT monitoring system, a three-method image interpolation comparison, and a measured LM386 audio amplifier with a calculated 200x gain — using Python, MATLAB, Verilog HDL, LTSpice, and AI-assisted workflows. Seeking AI-focused internship opportunities where I can apply practical engineering skills.

EXPERIENCE

FlyRank AI — AI Fluency Intern 2026

- Used AI tools including ChatGPT, Claude, Gemini, and Perplexity during AI Fluency assignments and workflows.
- Practiced prompt engineering by writing structured instructions, refining prompts, and constraining AI outputs to confirmed information.
- Audited workflows and explored how AI can support repetitive steps and improve workflow efficiency.

EDUCATION

Egypt-Japan University of Science and Technology (E-JUST) Expected 2028

B.Sc. in Computer Science & Engineering | School of ECCE | CGPA: 3.47 / 4.00

PROJECTS

IoT-Based Solar Panel Performance Monitoring System Arduino Uno · HC-06 · IoT Sensors

6-person team project, E-JUST coursework

- Contributed to a 6-person team building a monitoring system using an Arduino Uno, DS18B20 temperature sensor, GL10528 LDR, voltage sensor, shunt resistor, and I2C LCD.
- Used the HC-06 Bluetooth module to send data wirelessly to a phone or laptop.

Image Processing & Interpolation MATLAB · Python · OpenCV

Individual project, E-JUST coursework

- Implemented the MATLAB and Python portions of an image interpolation project, resizing images using Nearest Neighbour, Bilinear, and Bicubic interpolation methods.
- Compared the resulting images to examine differences in image quality across the three interpolation methods.

Audio Amplifier using LM386 IC Electronics Lab · LTspice

Individual project, ECE 212 – Electronics Engineering Lab

- Designed and built an audio amplifier circuit around the LM386 IC on a breadboard, and verified the design in LTspice before physical assembly.
- Calculated and confirmed a voltage gain of 200 (raised from the IC's default gain of 20 using a 22 µF capacitor between pins 1 and 8), matching an undistorted amplified waveform in LTspice simulation.
- Sized the output coupling capacitor to keep RMS output within the 8Ω/0.25W speaker's rating (calculated ≈ 1.41V RMS), preventing overdriving the speaker.

Quiz Engine Application Python

- Built an interactive quiz application with scoring, input validation, and a modular, object-oriented code structure for maintainability.

Engineering Your Future – Research Project Key Skills Course

- Conducted a survey-based study on factors influencing students' engineering-major selection; analyzed results and presented recommendations for improving career guidance.

CERTIFICATIONS

- AI Fluency: Framework & Foundations — Anthropic Education
- Claude 101 — Anthropic
- Introduction to Claude Cowork — Anthropic

TECHNICAL SKILLS

- AI & Data:** AI Tools, Prompt Engineering, Workflow Automation
- Programming & HDL:** Python, Verilog HDL (digital logic design)
- Tools & Platforms:** OpenCV, MATLAB, LTSpice (circuit simulation), SolidWorks (3D design), GitHub
- Professional Skills:** Technical Writing, Documentation, Problem Solving, Teamwork, Time Management
- Languages:** Arabic (Native), English (B2 – Upper Intermediate)

RELEVANT COURSEWORK

Digital Logic Design • Signals & Systems • Introduction to Electronics • Measurements & Instrumentation • Computer Programming
• Electrical Engineering Fundamentals • Mathematics for Engineering